



Faculty of Engineering & Technology
Department of Electrical & Computer Engineering
ENCS4310, DIGITAL SIGNAL PROCESSING (DSP)
Project Task 1,2
Correlation

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Section: 1

Date: 3/11/2024

is often used for pattern detection, and convolution is essential for system analysis.

Task 1:

What is the relation between convolution and correlation? (Discuss the expression and the use)

1. Concepts

- ❖ **Correlation:** measures the similarity between two signals by shifting one signal relative to the other and calculating the similarity at each shift. In discrete form, cross-correlation between two signals x and y is:

Equation1 :cross correlation between two signals x and y in discrete form

$$R_{x,y}(k) = \sum_{n=-\infty}^{\infty} x[n]y[n+k]$$

Where k represents the amount of shift applied to y relative to x .

- ❖ **Convolution:** is similar but includes a "flipping" of one signal before calculating similarity (before the shift).

Equation2 : The discrete convolution of x and y

$$(x * y)[n] = \sum_{m=-\infty}^{\infty} x[m]y[n-m]$$

This flipping operation makes convolution suited for systems analysis, where one signal is an input and the other represents the system's impulse response.

2. Expressions for Convolution and Correlation

- ❖ **Correlation:** Used primarily for comparing similarity between two signals.

Equation3 :Correlation simplify for finite sequences(time-limited sequences)

$$R_{x,y}(k) = \sum_{n=0}^{N-1} x[n]y[n+k]$$

- ❖ **Convolution:** Often used in signal processing to determine the output of a system given an input and the system's response. Unlike correlation, convolution involves flipping one of the sequences.

3. Use and Interpretation of Convolution and Correlation

Correlation and **convolution** are both fundamental operations in signal processing but serve different purposes. **Correlation** measures the similarity between two signals as one is shifted over the other without any flipping; it's used in applications like signal matching and detection to quantify how similar two signals are, such as identifying patterns or finding the location of one signal within another. **Convolution**, on the other hand, is used to determine how one signal (often an input) interacts with a system (represented by an impulse response) to produce an output. In convolution, one signal is flipped before the shifting, which makes it ideal for applications like filtering, where it calculates how an input signal is modified by a system.

4. Relation Between Convolution and Correlation

The primary difference between convolution and correlation lies in the flipping of one signal. In correlation, we slide one signal over the other directly, calculating similarity at each shift. In convolution, however, one signal is flipped before sliding, which changes how the signals combine, especially when analyzing time-based systems. Despite this difference, the two are mathematically related: **correlation** can be seen as a form of convolution without flipping one of the signals. This close relationship makes them complementary operations in signal processing, where correlation

5. Example Calculation

- ❖ **Given sequences:**

$$x = [1, 3, -2, 4]$$

$$y = [2, 3, -1, 3]$$

$$z = [2, -1, 4, -2]$$

- ❖ **Calculations:**

$R_{x,y}(k) = (1 \times 2) + (3 \times 3) + (-2 \times 1) + (4 \times 3) = 2 + 9 + 2 + 12 = 25$, the positive and relatively high value suggests a strong similarity between x and y

$R_{x,z}(k) = (2 \times 2) + (3 \times -1) + (-1 \times 4) + (3 \times -2) = 4 - 3 - 4 - 6 = -9$, The negative value suggests that x and z are less similar and possibly exhibit opposite patterns.

Since $R_{x,y}(k) > R_{x,z}(k)$, it indicates that y is more similar to x than z is to y .

However, standard correlation may not reliably identify similar sequences when one signal has a significantly different amplitude from the other. For instance, if we modify z to $w = [100, -1, 4, -2]$, the standard correlation $R_{y,w}$ might yield a high value which is 187 even though w and y are not truly similar. In such cases, **normalized correlation**—defined as the correlation coefficient $\sigma_{x,y}$ —is a better tool because it adjusts for amplitude differences.

The normalized correlation coefficient $\sigma_{x,y}$ is calculated by dividing the standard correlation by the product of the magnitudes (norms) of each signal:

$$\sigma_{x,y} = \frac{\sum_{n=0}^{N-1} x[n]y[n]}{\sqrt{\sum_{n=0}^{N-1} x[n]^2} \cdot \sqrt{\sum_{n=0}^{N-1} y[n]^2}}$$

For example, if we find that $\sigma_{x,y} = 0.95$ and $\sigma_{x,w} = 0.38$, the normalized correlation correctly identifies that x is more similar to y than w , despite the high standard correlation value of $R_{y,w}$. This shows that normalized correlation is more effective at identifying true similarity in the presence of amplitude differences.

6. Conclusion

-The correlation is used to measure the similarity between two signals without any flipping, useful for applications like pattern matching, while convolution involves a flipped signal, making it suitable for applications in system analysis.

-By understanding both standard and normalized correlation, we see that correlation is suitable for similarity detection but is best used in the normalized form for accurate results when amplitude differs across signals. Meanwhile, convolution serves different applications, such as filtering and system response analysis, where one signal is flipped before shifting and combining.

Task 2:

check which one (standard or normalized) is more beneficial to identify how strongly one signal is present in two different signals. (Matlab)

Let $S_i = \cos(2\pi f_i t)$

s_1 : 1 Hz, s_2 : 4 Hz, s_3 : 10 Hz

The following signals contain the three sinusoids above

$X = 2S_1 + 4S_2 + S_3$

$Y = S_1 + S_2$

a) Plot the generated signals for 1 sec. let the sampling interval 0.01 sec

b) Compare which is useful for identifying how strongly one signal (say S_1) is present in another.

1. Define Correlation Types

- **Standard Correlation:** Measures similarity without adjusting for amplitude. It gives a raw similarity score, which may vary significantly depending on the amplitude of the signals.
- **Normalized Correlation:** Adjusts for amplitude by dividing by the magnitude of the signals, providing a similarity measure that is independent of signal strength.

2. Compute Correlation in MATLAB:

```
1 % Parameters
2 f1 = 1; % Frequency of S1 in Hz
3 f2 = 4; % Frequency of S2 in Hz
4 f3 = 10; % Frequency of S3 in Hz
5 t = 0:0.01:1; % Time vector from 0 to 1 second with a sampling interval of 0.01
6
7 % Define the signals S1, S2, and S3
8 S1 = cos(2 * pi * f1 * t);
9 S2 = cos(2 * pi * f2 * t);
10 S3 = cos(2 * pi * f3 * t);
11
12 % Define the composite signals X and Y
13 X = 2 * S1 + 4 * S2 + S3;
14 Y = S1 + S2;
15
16 % Compute cross-correlation between X and S1, and between Y and S1
17 [corr_X_S1, lags_X] = xcorr(X, S1, 'biased');
18 [corr_Y_S1, lags_Y] = xcorr(Y, S1, 'biased');
19
20 % Normalize the cross-correlations
21 norm_corr_X_S1 = corr_X_S1 / max(abs(corr_X_S1));
22 norm_corr_Y_S1 = corr_Y_S1 / max(abs(corr_Y_S1));
23
24
25 % Plot all figures in a single figure using subplots
26 figure;
27
28 % Plot Signal X
29 subplot(3, 2, 1);
30 plot(t, X);
31 title('Signal X = 2S1 + 4S2 + S3');
32 xlabel('Time (s)');
33 ylabel('Amplitude');
34
35 % Plot Signal Y
36 subplot(3, 2, 2);
37 plot(t, Y);
38 title('Signal Y = S1 + S2');
39 xlabel('Time (s)');
40 ylabel('Amplitude');
41
42 % Plot Reference Signal S1
43 subplot(3, 2, 3);
44 plot(t, S1);
45 title('Reference Signal S1');
46 xlabel('Time (s)');
47 ylabel('Amplitude');
48
49
50 % Plot Standard and Normalized Cross-Correlation between X and S1
51 subplot(3, 2, 4);
52 plot(lags_X * 0.01, corr_X_S1, 'b', 'DisplayNone', 'Standard');
53 hold on;
54 plot(lags_X * 0.01, norm_corr_X_S1, 'r--', 'DisplayNone', 'Normalized');
55 title('Cross-Correlation between X and S1');
56 xlabel('Lag (s)');
57 ylabel('Correlation');
58 legend;
59 hold off;
60
61 % Plot Standard and Normalized Cross-Correlation between Y and S1
62 subplot(3, 2, 5);
63 plot(lags_Y * 0.01, corr_Y_S1, 'b', 'DisplayNone', 'Standard');
64 hold on;
65 plot(lags_Y * 0.01, norm_corr_Y_S1, 'r--', 'DisplayNone', 'Normalized');
66 title('Cross-Correlation between Y and S1');
67 xlabel('Lag (s)');
68 ylabel('Correlation');
69 legend;
70 hold off;
```

Figure 1 : The MATLAB code

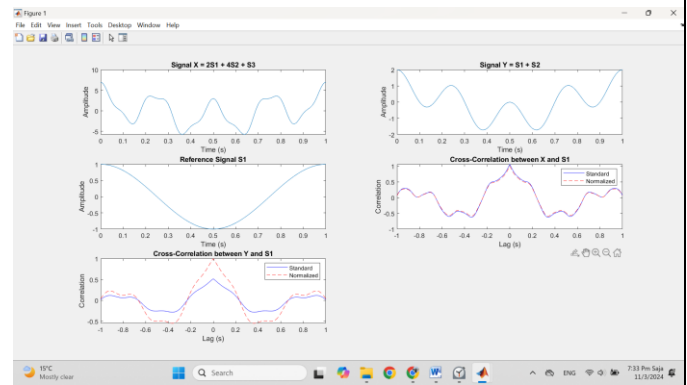


Figure 2: The result from the code

Normalized cross-correlation is more useful for identifying how strongly one signal (like S_1) is present in another signal (like X or Y).

Reason:

Normalized cross-correlation scales the correlation values between -1 and 1, making it easier to identify the similarity in pattern or presence of one signal within another without being influenced by differences in amplitude. This makes it clearer to detect the presence of S_1 within X and Y , even if other components with higher amplitudes (like S_2 and S_3) are also present in the signals.